

CY 504: Heterocyclic Chemistry

Course Code	CY 504
Course Name	Heterocyclic Chemistry
L-T-P-C	2-0-0-2
Prerequisites	CY501
Intended for	UG/PG
Distribution	Core course for M.Sc Chemistry
Approval	9th Senate (May 2016)

Course Contents

- **Module I**
Nomenclature, Classification of heterocycles; Common synthesis strategies for aromatic heterocycles; Furan, Pyrrole and Thiophene (five membered heterocycles with one heteroatom) general methods and strategies for their synthesis; their properties, structure and reactivity; their electrophilic substitution and metalation. [7 Lectures].
- **Module II**
Five membered heterocycles with two heteroatoms- Pyrazole, Imidazoles, Thiazole, Isothiazole; Isoxazole and Oxazole - Brief introduction, synthetic approaches and chemical reactions. [5 Lectures].
- **Module III**
Six membered heterocycles with one heteroatom. Pyridines general properties, structure, synthesis strategies, electrophilic and nucleophilic substitution reactions in pyridines, metalation of pyridines; Pyridine derivatives such as oxy-pyridines, alkyl pyridines, pyridinium salts, and pyridine N- oxides [5 Lectures]
- **Module IV**
Fused heterocyclic systems. Indole - electronic structure and reactivity of indoles, Fisher and Bischler indole syntheses, reaction of indole with electrophiles, Mannich reaction of Indoles; Quinoline and Isoquinoline - General properties and reactivity, synthesis strategies, electrophilic and nucleophilic substitution reactions in quinolones and Isoquinolines. [7 Lectures].
Brief on the modern methods and applications.
- **Module 5**
Six membered heterocycles with two heteroatoms. Pyridazine, Pyrimidine, Pryazine, and Oxazines brief introduction, synthesis strategies and reactions. [4 Lectures]

Text Books

1. J. A. Joule, K. Mills and G. F. Smith, **Heterocyclic Chemistry**, 3rd Edition.
2. T. L. Gilchrist, **Heterocyclic Chemistry**, 2nd Edition.

References

1. D. T. Davies, **Aromatic Heterocyclic Chemistry**
2. T. Eicher and S. Hauptmann, **The Chemistry of Heterocycles**.
3. Lee A. Paquette, **Principles of Modern Heterocyclic Chemistry**
4. P-de-Mayo, **Molecular Rearrangement**.